

Assignment 1

1. Explain Computer Science and its major components. How does it differ from Information Technology?
2. Define an algorithm. Explain its characteristics with a real-life example.
3. Explain data representation in computers. Discuss binary representation and its importance.
4. Differentiate between hardware, software, and operating system with suitable examples.
5. Explain the role of an operating system in managing computer resources.
6. Explain the steps to install Python and run a program using IDLE.
7. Define variables and data types in Python. Explain immutable and mutable types with examples.
8. Explain numerical data types and operators in Python with examples.
9. What are expressions in Python? Explain different types with examples.
10. Explain the importance of indentation and comments in Python programming.
11. Explain Boolean logic and logical operators with truth tables.
12. Write a Python program using if-else and nested if-else to find the largest of three numbers.
13. Explain break, continue, and pass with suitable code examples.